

Small Office Network – Static & Dynamic Topology (Packet Tracer)

Project Overview

This lab implements a small office network divided into two segments:

- **Static Network (192.168.10.0/24)**
- **Dynamic Network with DHCP (192.168.20.0/24)**

Topology Summary

The topology consists of two independent networks connected through routers.

Static Network

- Cisco **1941 Router (R1)**
- Cisco **2960 Switch**
- 4 PCs (PC0–PC3)
- 1 Server
- All devices configured with **static IP addresses**
- Dual-stack IPv4 & IPv6

Dynamic Network

- Cisco **1941 Router (R2)**
- Cisco **2960 Switch**
- 2 PCs (PC4–PC5)
- Access Point
- Laptop connected via WiFi
- **DHCP** enabled on the router

Interconnection

- A direct link between **R1** and **R2** allows communication between both subnets.

IP Addressing

Static Network

- **Subnet:** 192.168.10.0/24
- **Default Gateway:** 192.168.10.1
- Manual IP configuration on all hosts

Dynamic Network

- **Subnet:** 192.168.20.0/24
- **DHCP Gateway:** 192.168.20.1
- Router assigns IPs to PCs and laptop

IP Addressing Table

Device	IP Address	Subnet Mask	Default Gateway
R1 G0/0	192.168.10.1 2001:db8:10::1/64	255.255.255.0	N/A
R1 G0/1	10.10.10.1	255.255.255.252	N/A
R2 G0/0	192.168.20.1	255.255.255.0	N/A
R2 G0/1	10.10.10.2	255.255.255.252	N/A
PC0	192.168.10.10 2001:DB8:10::10/64	255.255.255.0	192.168.10.1
PC1	192.168.10.11 2001:DB8:10::11/64	255.255.255.0	192.168.10.1
PC2	192.168.10.12 2001:DB8:10::12/64	255.255.255.0	192.168.10.1
PC3	192.168.10.13 2001:DB8:10::13/64	255.255.255.0	192.168.10.1
PC4	DHCP	255.255.255.0	192.168.20.1
PC5	DHCP	255.255.255.0	192.168.20.1
Laptop	DHCP	255.255.255.0	192.168.20.1

Server	192.168.10.100 2001:DB8:10::100/64	255.255.255.0	192.168.10.1
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Configuration Details

Router R1 (Static Network)

- Gig0/0 and Gig0/1 configured
- Default gateway for hosts
- Static route to reach 192.168.20.0/24
- SSH enabled
- Local admin user configured
- RSA keys generated

Router R2 (Dynamic Network)

- Gig0/0 and Gig0/1 configured
- Default gateway for hosts
- DHCP pool for 192.168.20.0/24
- Static route to reach 192.168.10.0/24
- SSH enabled
- Local admin user configured
- RSA keys generated

Switches

- Default VLAN
- Connections to routers and end devices

Access Point

- SSID configured
- WPA2 security
- Laptop successfully connected

Server

- DNS configured
- HTTP configured

Cabling

- Straight-through cables to connect hosts to the switch
- Crossover cable to connect both routers

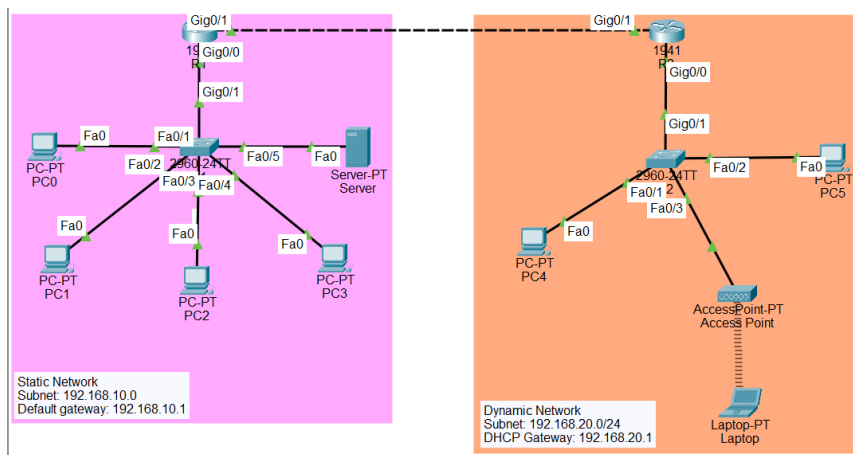
Testing & Verification

- Ping tests within each subnet
- Ping tests across subnets (inter-network routing)
- Tracert verification test
- DHCP assignment verification
- Wireless connectivity test
- Server access from both networks

Included Files

- `SmallOfficeNetwork.pkt` – Packet Tracer project file
- `README.md` – Documentation
- PDF – Documentation

Screenshots



R1

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Static Routes

Network

Mask

Next Hop

Add

Network Address

192.168.20.0/24 via 10.10.10.2

R2

PhysicalConfigCLIAttributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

Static Routes

Network

Mask

Next Hop

Add

Network Address

192.168.10.0/24 via 10.10.10.1

Laptop

PhysicalConfigDesktopProgrammingAttributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

Wireless0

Bluetooth

Wireless0

Port Status

Bandwidth

MAC Address

SSID

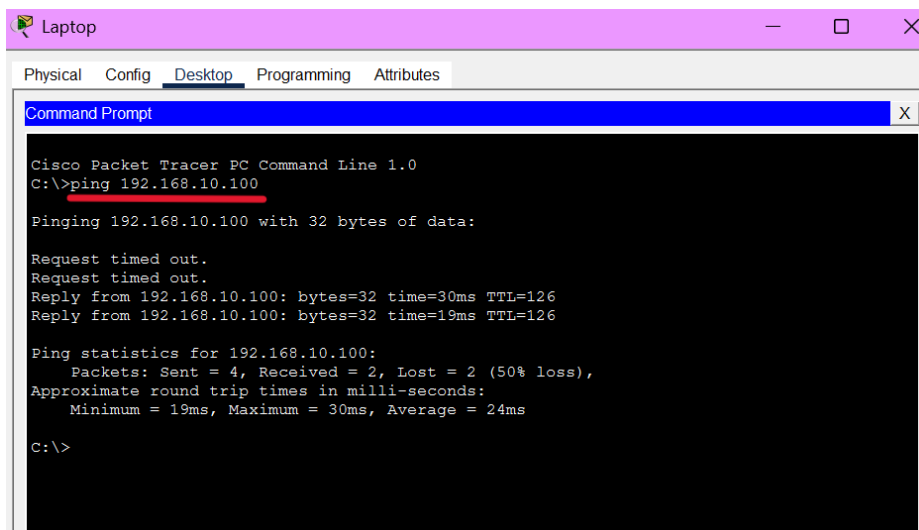
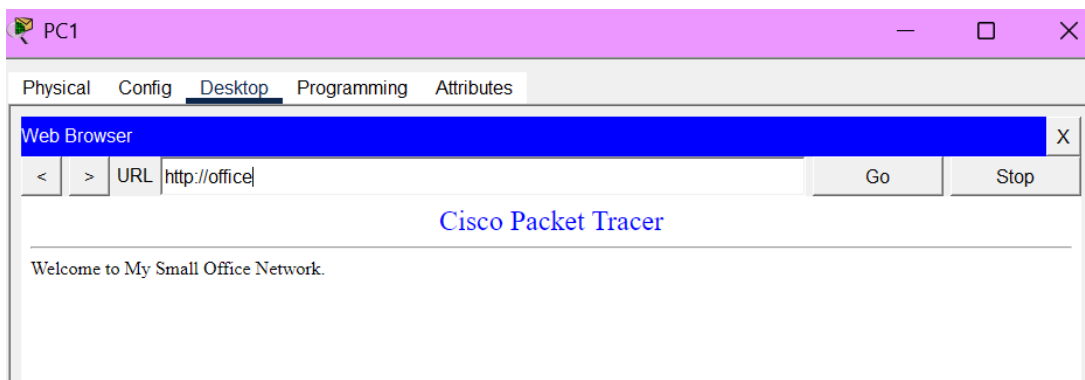
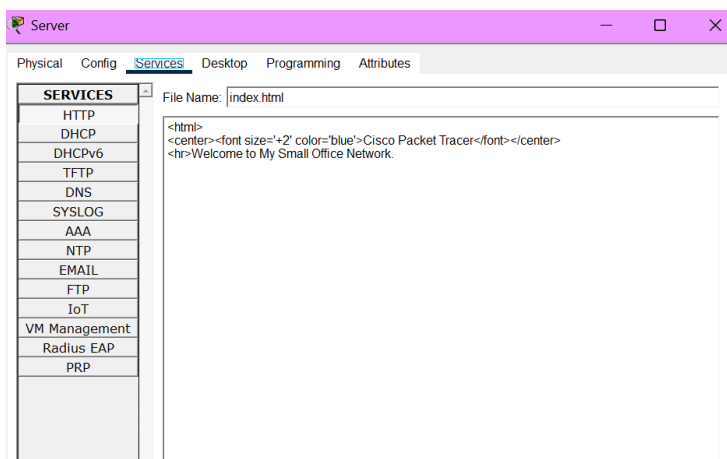
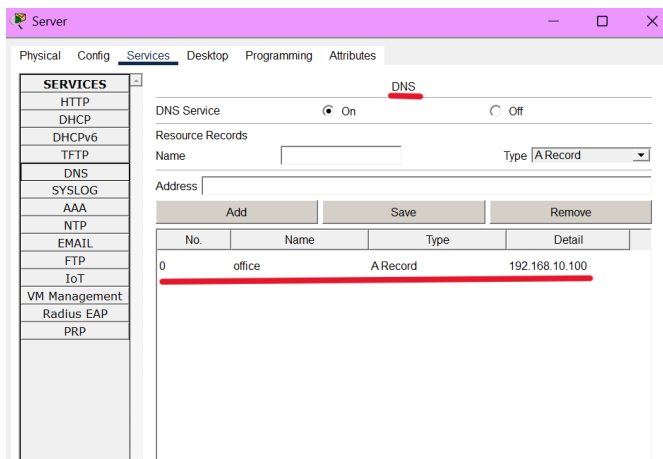
Authentication

Encryption Type

IP Configuration

IPv6 Configuration

Link Local Address



PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>tracert 192.168.20.2

Tracing route to 192.168.20.2 over a maximum of 30 hops:

  1  0 ms    3 ms    3 ms    192.168.10.1
  2  3 ms    13 ms   0 ms    10.10.10.2
  3  *       13 ms   13 ms   192.168.20.2

Trace complete.

C:\>
```

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 10.10.10.1

Password:

R1>enable
Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
```

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...: 
    Physical Address...: 0001.42D5.7321
    Link-local IPv6 Address...: FE80::201:42FF:FED5:7321
    IPv6 Address...: 2001:DB8:10::12
    IPv4 Address...: 192.168.10.12
    Subnet Mask...: 255.255.255.0
    Default Gateway...: ::
                        192.168.10.1
    DHCP Servers...: 0.0.0.0
    DHCPv6 IAID...: 
    DHCPv6 Client DUID...: 00-01-00-01-B8-E1-AE-34-00-01-42-D5-73-21
    DNS Servers...: ::
                    0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address...: 0090.2BC1.4DEB
    Link-local IPv6 Address...: ::
--More--
```